



The role of artificial intelligence (AI) in architectural design: a systematic review of emerging technologies and applications

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Abstract

Recent advancements in Artificial Intelligence (AI) are reshaping architectural design by enhancing key design techniques such as spatial planning, parametric modeling, generative design, and performance-based analysis. AI methods like machine learning (ML) and predictive modeling support tasks from material selection to structural optimization, improving efficiency, sustainability, and creativity. By analyzing large datasets and automating complex processes, AI empowers architects to explore innovative solutions responsive to user needs and environmental factors. This study systematically reviews how AI plays a part in the architectural workflow, focusing on its ability to enhance creativity, automation, and sustainability. A structured literature review was conducted using the PRISMA framework to review peer-reviewed studies on AI applications in architectural design, urban planning, and smart cities published between 2003 and 2025. The reviewed studies demonstrate that AI enhances generative design, streamlines spatial organization, and supports sustainable architecture. However, challenges such as algorithmic bias, ethical concerns, and loss of architectural identity persist. Future research should emphasize ethical implementation, interdisciplinary collaboration, and integration with VR/AR to enable immersive, real-time, and informed design processes.

Keywords Artificial intelligence (AI) · Augmented reality · Architectural design · Technologies

Abbreviations

3DBP	3D Building Printing	DRL	Deep Reinforcement Learning
ACO	Ant Colony Optimization	DSS	Decision-Making Support Tools
AF	Additive Fabrication	ES	Knowledge-Based Systems
AI	Artificial Intelligence	FNN	Feedforward Neural Network
AL	Automated Learning	NNs	Feedforward Neural Networks
AR	Augmented Reality	GAN	Generative Adversarial Networks
BEM	Building Energy Modeling	GD	Generative Draft
BIM	Building Information Modeling	GIS	Geographic Information System
CAD	Computer-Aided Design	GWO	Grey Wolf Optimizer
CFD	Computational Fluid Dynamics	HVAC	Heating, Ventilation, and Air Conditioning (Innovative)
CNA	Convolutional Neural Architectures	ML	Machine Learning
CNN	Convolutional Neural Networks	PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
DBMS	Database Management Frameworks	PSS	Planning Assistance Platforms
DI	Digital Twin	R-CNN	Region-Based CNN
DL	Deep Learning	RL	Reinforcement Learning
		SDG	Sustainable Development Goals
		SVM	Support Vector Machines
		VDC	Virtual Design and Construction
		VR	Virtual Reality

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1 Introduction

Artificial Intelligence (AI) is increasingly redefining architectural design by introducing data-driven methods that enhance efficiency, precision, and spatial adaptability [1]. Generative design is among its most transformative applications, where algorithms autonomously generate and evaluate thousands of design alternatives based on defined constraints such as form, function, materiality, and energy performance [2]. These tools offer a departure from intuition-based practices, allowing architects to balance structural integrity, sustainability, and spatial innovation through computational precision [1–4].

Machine learning (ML) and neural networks have transformed architectural planning by replacing intuition-based decisions with data-driven insights, optimizing space, predicting performance, and enhancing energy efficiency [5, 6]. The increasing adoption of ML across various architectural applications highlights its potential to improve design accuracy, sustainability, and structural integrity [5, 6].

Despite its potential, the integration of computational intelligence in architectural planning remains constrained. Automated learning (AL) has been applied to design space exploration, blueprint generation, style adaptation, and environmental evaluation, but data availability challenges hinder its broader implementation [7]. While the AL approach can enhance the assessment of ecological efficiency for architects, it has been proposed that AL could be a powerful tool in significantly lowering the carbon footprint of structures. However, incorporating AL in architectural workflows is restricted by a fundamental challenge—data availability. These computational models necessitate vast volumes of annotated input data to identify underlying trends within datasets [5, 8]. For example, an extensive collection of previously computed cases is essential to develop an algorithm capable of accurately forecasting the lifecycle carbon emissions of a planned structure. Ideally, this dataset should be sourced from measurements to ensure maximum accuracy. Nevertheless, in many scenarios, obtaining such data is not feasible. Regarding carbon footprint evaluation, it is possible to digitally model various architectural forms, analyze their embodied and operational carbon, and utilize these instances to train the AL system [5, 8, 9].

Several studies have reviewed the current applications and future directions for improving energy efficiency, emphasizing its effectiveness in homes and its role in fostering more intelligent, sustainable environments while reducing carbon emissions [10, 11]. From a broader perspective, a study by Alnaser et al. (2024) investigated the concept of a green-built environment, which incorporates eco-friendly and resource-efficient strategies throughout a structure's entire lifespan, from conceptualization to deconstruction

[12]. Furthermore, Alnaser et al. (2024) and Ilvitskaya et al. (2018) explore how these environments reduce ecological footprints by enhancing energy conservation, minimizing material waste, and utilizing sustainable resources while improving occupant health and comfort [9, 12].

Despite global efforts, challenges persist due to the underutilization of new technologies, financial constraints, and inconsistent regulations that governments face. Many studies support the UN Sustainable Development Goals (SDGs) concerning clean energy, sustainable cities, and climate action [13, 14]. This research underscores the potential of emerging technologies, including AI and ML, in enhancing energy efficiency. When used together, the simulation tools Design Builder and EnergyPlus help architecture, engineering, and research professionals achieve optimal building designs while meeting energy efficiency and sustainability goals [15].

Data-driven strategies and computational techniques are transforming the construction sector by enhancing productivity through automated planning, digital infrastructure, simulations, and structural optimization [16].

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In parallel, the global construction industry faces ongoing challenges including declining productivity, workforce shortages, and rising project complexity. In response, AI and automation have been proposed as solutions within the framework of the Fourth Industrial Revolution. These include AI-driven structural optimization, automated design generation, and robotic construction systems, each aiming to streamline workflows, improve resource allocation, and address skill gaps across the sector [16]. Furthermore, AI's influence transcends architecture, as evidenced by landmark achievements in other fields. For instance, AlphaGo's disruption of traditional game theory reflects the broader capacity of AI to recalibrate human decision-making and strategic planning. In architecture, similar paradigm shifts are unfolding as AI transitions from a supportive tool to an active agent in creative and technical processes [16, 17].

In light of these developments, this systematic review seeks to consolidate emerging research on AI in architectural design. By critically examining the technologies,

applications, challenges, and future directions, this study aims to provide a coherent synthesis that informs both academic inquiry and professional practice in the era of computational design.

1.1 Research problem

In recent years, architectural design has entered a transformative phase marked by the integration of artificial intelligence (AI) across planning, modeling, and construction workflows. Although multiple technologies have been introduced and theoretical contributions continue to expand, a fragmented understanding persists regarding the systematic role of AI in addressing contemporary architectural challenges such as energy performance, spatial optimization, and creative automation. Disciplinary fragmentation, software incompatibility, and educational deficiencies further complicate the development of a unified AI design framework. As a result, there remains an urgent need to consolidate technological advances, practical applications, and interdisciplinary insights to evaluate how AI can be effectively operationalized within the built environment.

1.2 Research questions

To achieve these objectives, the following research questions are addressed:

- What are the most significant AI technologies currently utilized in architectural design, and how do they contribute to the field's evolution?
- In what ways does AI enhance creativity and facilitate more effective decision-making in architectural processes?
- What are the primary challenges associated with AI-driven architecture, and how can these obstacles be addressed to maximize AI's potential in the industry?
- How is AI expected to influence the future of architectural design, and what emerging trends will shape its development in the coming years?

How is artificial intelligence currently applied in architectural design, and in what ways do these technologies enhance generative creativity, environmental performance, and interdisciplinary integration within contemporary design practice? By addressing these questions, this review aims to provide a comprehensive overview of how AI is reshaping architectural design and offer insights into this technology's potential future directions.

1.3 Main aim

This study aims to systematically examine and synthesize the emerging applications of artificial intelligence in architectural design, with particular attention to how such technologies influence creativity, sustainability, and professional collaboration across disciplinary boundaries.

1.4 Research objectives

This study seeks to comprehensively examine the role of AI in contemporary architectural design by systematically reviewing prevailing technologies and their practical implementations. The primary objectives are as follows:

- To identify and categorize the principal AI methodologies, including machine learning, deep learning, neural networks, and generative design, currently employed within architectural design and planning processes.
- To investigate how AI augments fundamental architectural design practices, such as spatial configuration, material optimization, structural analysis, energy efficiency modeling, and parametric design strategies.
- To assess the convergence of AI with emerging digital technologies, notably virtual reality (VR), augmented reality (AR), 3D printing, and building information modeling (BIM), and its influence on architectural visualization, design automation, and construction workflows.
- To critically evaluate the technical constraints and implementation challenges associated with the adoption of AI in architectural contexts, including issues related to data availability, algorithmic complexity, and system integration.
- To examine the technical, pedagogical, and ethical challenges limiting the broader adoption of AI in architecture.
- To propose future research directions that respond to current knowledge gaps and promote inclusive, collaborative frameworks for AI integration in architecture.

1.5 The evolution of AI in architectural design

Since ancient times, people have tried to create systems that mimic how the human brain works. The implementation of AI has evolved logically over time through continuous advancements in mathematical reasoning, physical sciences, and philosophical thought, alongside rapid progress in computational technology [18]. Computational intelligence encompasses a broad field that examines and enhances human thinking through its fundamental principles and technology-driven components. AI systems have been continuously developing for more than sixty years before

achieving widespread application across various domains, with architectural planning being one of the primary areas of implementation. Computational intelligence started as a design tool in architecture but has become a partner for architects. It could lead to the decision-making process [19].

Architecture has a long history of computational innovation, from the earliest computer-aided design (CAD) software to the current use of parametric design tools. Nevertheless, the actual revolutionization of architectural workflows came due to the advent of ML and deep learning (DL) technologies [20]. Unlike rule-based systems, AI algorithms can learn from vast datasets and generate new design solutions by identifying patterns and trends, making them dynamic and versatile. A particular form of AI, generative adversarial networks (GANs), has been widely popular in architecture because they can generate new design concepts with human-like creativity [21]. The shift to AI-powered methodologies for architectural design is quite profound, and this means moving from static computational tools to them. For instance, using historical data, ML is utilized in predictive modeling with AI systems predicting future design needs or problems in a particular field. Additionally, AI can automatically optimize building layouts and configurations to maximize space utilization and energy consumption, providing more terrific design refinement that is not feasible with traditional methods [22].

2 Research methodology

2.1 Inclusion and exclusion criteria

For this review, the studies were selected based on a rigorous set of inclusion and exclusion criteria to ensure relevance and quality. The search terms included “AI in architecture,” “generative design in architecture,” “ML in urban planning,” “neural networks for sustainable architecture,” “AI-driven smart cities,” and “automation in architectural processes.”

The inclusion criteria prioritized peer-reviewed journal articles, conference proceedings, and case studies that offered substantial contributions to understanding AI in architectural design. The selected studies were published between 2003 and 2025, a period marked by significant advancements in AI-driven methodologies. The review focused on research applying AI to architectural design, including Building Information Modeling (BIM) and sustainable construction. Empirical studies and real-world case studies were included to provide concrete evidence of AI’s practical impact.

On the contrary, studies that did not align with the objectives of this review were excluded. Computational design

methods remained relatively outdated before the rapid advancement of AI. Studies focused solely on theoretical AI models without practical applications were excluded, as this review emphasizes real-world implementations. Research unrelated to architecture—such as work in mechanical engineering or robotics—was also omitted. Additionally, papers lacking sufficient experimental validation were excluded to ensure credibility. These criteria narrowed the review to two high-quality studies demonstrating robust evidence of AI-driven architectural processes.

2.2 Study selection

This review adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework to ensure transparency and reproducibility in study selection. The PRISMA flow diagram (Fig. 1) delineates the stepwise filtration process, beginning with the identification of 1,500 records across databases such as Web of Science, Scopus, and Google Scholar. Following title and abstract screening, 450 full-text articles were evaluated against defined inclusion criteria. Studies were selected based on their empirical application of AI in architectural design, publication between 2003 and 2024, and relevance to generative, sustainable, or urban-scale design. Exclusion criteria included theoretical-only papers, non-English language publications, and research not specific to architecture. Data extraction followed a structured protocol encompassing AI methods employed, performance metrics, application domains, and outcome relevance. Quality assessment was conducted using journal ranking indicators, peer-review status, and evidence of experimental validation, ensuring that only high-impact, methodologically sound studies informed the synthesis.

3 Literature review

3.1 Core AI technologies in architectural practice

Machine Learning (ML) encompasses three main categories: supervised, unsupervised, and reinforcement learning. In supervised learning, algorithms perform classification (e.g., decision trees, support vector machines (SVMs) and regression (e.g., linear regression, neural networks), recognizing patterns and making informed predictions based on labeled datasets [23]. Unsupervised learning, by contrast, identifies patterns in unlabeled data using clustering techniques such as centroid-based, mean-shift, and probabilistic clustering models. Reinforcement learning optimizes decision-making through trial-and-error approaches [23].

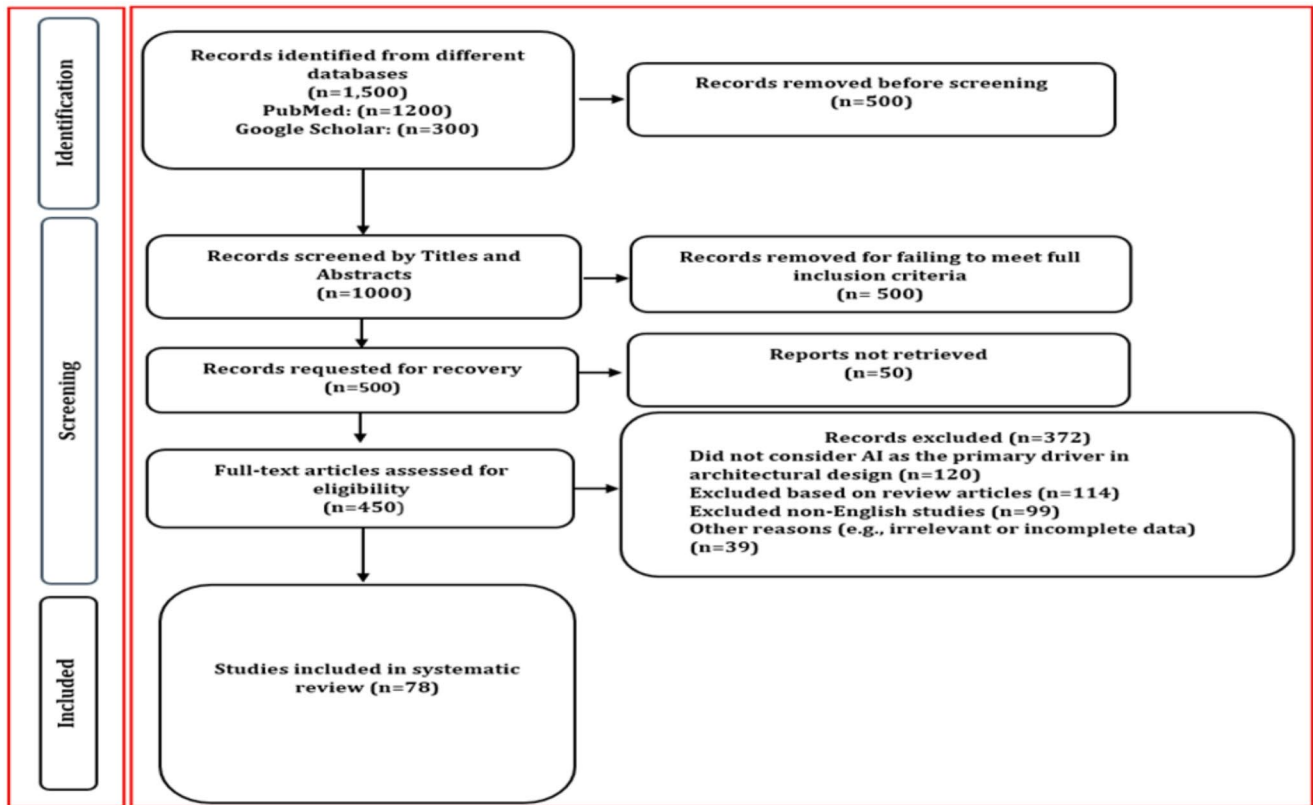


Fig. 1 PRISMA Flow Diagram

These ML techniques have been widely implemented in intelligent energy management systems, significantly enhancing energy efficiency in buildings [24]. Researchers have demonstrated that ML algorithms optimize energy consumption, contributing to improved sustainability and cost-effectiveness in architectural projects. Automated learning (AL), a branch of computational intelligence, relies on statistical principles rather than explicit programming. It identifies recurring trends within extensive datasets and develops mathematical frameworks representing data dynamics. Different categories of AL exist; some rely on predefined training datasets, while others acquire knowledge independently, without human intervention. Neural architectures, particularly Convolutional Neural Architectures (CNA) [8], use convolutional layers to enhance visual data analysis, processing images through neural networks with input, output, and hidden layers [8, 25].

Despite its potential, AL integration in architectural planning is constrained by data availability challenges. AL has been applied to design space exploration, blueprint generation, style adaptation, and environmental evaluation [7]. However, these computational models require vast volumes of annotated data to identify underlying trends accurately [8]. For instance, predicting lifecycle carbon emissions requires an extensive collection of previously computed

cases, ideally based on empirical measurements. In many scenarios, such data is not readily available. In some cases, digital modeling of architectural forms and the analysis of their embodied and operational carbon can create the data needed to train AL systems [8, 9].

According to various studies [26, 27] comparing different assessment and certification systems, the categories that consistently carry the highest weights include environmental loading, internal environment, external environmental impact, and resource use. Generally, environmental factors are considered more relevant than socioeconomic and political aspects, as well as planning and innovation [26, 27].

The weightings of these systems are determined by the degree of importance assigned to each category of evaluation, and these priorities are directly shaped by the specific contexts and challenges of each country. Among the various systems, energy efficiency requirements consistently receive higher weighting, reflecting its recognition as one of the most urgent needs in sustainable architectural design. Other high-priority categories include material sustainability, potable water conservation, and parameters that influence indoor environmental comfort, such as visual, thermal, and acoustic comfort, as well as indoor air quality [26–28].

Based on these weightings and taking into account the architects' thought process during design, five key

sustainability indicators were identified: energy efficiency, potable water conservation, indoor air quality, acoustic comfort, and project durability. These indicators were deemed the most representative and suitable for integration into the methodological framework of architectural design practices [29].

Numerous studies highlight the contributions of AI to architectural optimization. Chew et al. (2024) applied deep learning (DL) techniques to predict structural integrity, significantly reducing design iteration time [30]. Sharma et al. (2024) explored the use of ML in automated zoning analysis, demonstrating its potential to enhance the efficiency of architectural design [31]. AI has also proven valuable in space utilization, rapidly generating new spatial layouts based on thousands of existing layouts [32]. These methods can theoretically solve spatial utilization problems faster than manual design. Nonetheless, challenges such as interpretability and data bias persist, necessitating human oversight in AI-driven design [32].

Building performance predictions increasingly rely on predictive modeling that incorporates climatic, structural, and user behavioral data [33]. AI-driven methods for space optimization enhance efficiency and improve spatial layout materiality, while adaptive building structures enable architects to design dynamic frameworks that respond to environmental changes [34]. These advancements collectively contribute to the evolution of intelligent and responsive architectural designs [34].

Rodrigues et al. [35] discuss the incorporation of DL with 3D building modeling and Building Information Modeling (BIM) to automate structural degradation classification in historic buildings. They use a region with CNN features (R-CNN) that has been trained using images that outline architectural anomalies. At the same time, a laser scanning process creates a point cloud reference to support the Building Information Modeling (BIM) model. This new approach automates the detection and classification of structural faults, relating the geometric and numerical information with BIM to support facility management [35]. The workflow has four significant steps: data acquisition through terrestrial laser scanning, modeling in 3D using BIM, image processing for DL-based defect identification, and incorporation of these results into Revit software. The research highlights the ability of digital technologies and AI to update conservation activities and deliver a structured decision-support system for architectural heritage upkeep. The primary challenge met is the automatic detection of anomalies, previously conducted manually, that ensures more trustworthy and transparent documentation within a digital ecosystem based on BIM [35].

Furthermore, ML techniques have been applied to sustainable building materials. Wang et al. [36] present a study

on leveraging ML optimization to predict the compressive strength of eco-friendly concrete or geopolymers concrete, an environmentally sustainable alternative to traditional cement-based concrete. The research utilizes a feedforward neural network (FNN) enhanced by the bird-inspired optimization technique, the Grey Wolf Optimizer (GWO), to refine predictive accuracy [36]. These studies underscore AI's multifaceted role in advancing architectural planning, construction, and heritage conservation.

A comprehensive summary of the key AI technologies, their applications, challenges, and future prospects in architectural design and planning is presented in Table 1 below.

3.2 Generative design and computational creativity in architecture design and planning

Generative design (GD) and computational creativity are increasingly transforming architectural practice by leveraging data-driven strategies to improve productivity through automated planning, digital infrastructure, simulations, and structural optimization [16]. Recent research has focused on generative modeling, which minimizes manual labor and time by automatically generating diverse design alternatives. This approach streamlines repetitive tasks, enhances operational efficiency, and supports industry professionals by using algorithmic frameworks to deliver practical, smart, and customized solutions [37].

The reviewed literature includes studies examining Autodesk Generative Design's impact on parametric architecture [38]. According to a study [39], AI has already been used in adaptive facades that minimize energy consumption through real-time environmental data analysis. Despite these advancements, AI-generated designs often lack intuitive human creativity, necessitating architectural refinement and human interpretation before implementation [39].

GD has rapidly gained traction in architectural research, reflecting its growing role in academic and professional practice [27]. GD involves creating multiple design options, evaluating their performance, and refining them through optimization. Marsh (2008) focused on GD's performance, while other researchers emphasized the process of generating, evaluating, and iteratively improving design options. These studies often classify GD according to either performance outcomes or process methodologies. In this context, GD development is typically analyzed based on algorithms and design constraints [29]. Algorithms act as programming instructions for creating and refining design alternatives. Common approaches include evolutionary techniques, shape-based design rules, fractal patterns, swarm intelligence, and cellular growth models. The selection of an algorithm depends on the project goal, with shape-based rules suited for space layout exploration and evolutionary

Table 1 AI technologies and its applications in architecture design and planning

AI technology	Key application & Advantages	Challenges	Functionality	Real-world applications	Future prospects	Ref.
Machine Learning (ML)	Predictive modeling enhances energy efficiency, reduces costs, and improves urban planning through energy consumption forecasting, structural safety assessments, automated zoning analysis, and sustainable material optimization.	It requires extensive, high-quality datasets, but there is potential data bias.	Analyzes large datasets to identify patterns and make data-driven decisions for architectural design and urban planning.	Used in smart energy grids to optimize HVAC systems and predict material lifespan.	Expected to improve automation in sustainable construction and intelligent building management.	[24, 30–32, 34, 36]
Neural Networks (NNs)	Analysis of smart city infrastructure, lifecycle carbon emissions forecasting, and predictive modeling for building performance. Enhances automation and data-driven decision-making.	High computational demand requires extensive labeled data.	Processes complex, high-dimensional data to identify correlations between architectural factors.	Applied in carbon footprint analysis of buildings and climate-responsive architectural designs.	Future developments may lead to real-time carbon tracking in urban environments.	[8, 9, 12]
Deep Learning (DL)	Predicting structural integrity, classifying historic building defects, and integrating BIM improve design efficiency and structural analysis accuracy.	Computationally expensive and requires significant training data.	Uses multi-layered neural networks to analyze architectural projects' images, structures, and materials.	Employed in automated fault detection in heritage building conservation.	Expected to enhance predictive maintenance and AI-driven architectural restoration.	[30, 31, 35, 36]
Automated Learning (AL)	Design space exploration, environmental evaluation, energy efficiency analysis, and blueprint generation. Reduces carbon footprint and enhances ecological efficiency.	It requires vast annotated datasets, but data availability has limitations.	Learns from past design iterations and optimizes energy-efficient architecture.	Utilized in smart city projects to improve urban infrastructure design.	Future AL systems may autonomously generate green building plans.	[8, 9]
Convolutional Neural Networks (CNNs)	Visual data analysis, defect detection, and classification in historic buildings. High accuracy in pattern recognition and predictive maintenance.	Requires large image datasets and computational power.	Extracts features from architectural images for pattern identification.	Used in drone-based structural damage assessment.	Integration with AR for real-time defect detection.	[8, 9, 35]
Reinforcement Learning (RL)	Adaptive zoning regulations, land-use optimization, and AI-driven space optimization. Improves efficiency in land allocation and regulatory compliance.	Computationally intensive; requires complex simulation environments.	Uses trial-and-error learning to optimize land-use planning.	Implemented smart zoning tools for urban development.	Future AI-driven land-use models may predict and prevent overpopulation stress.	[31–33]
Generative AI	AI-driven space layout optimization based on existing designs. Faster design iterations and optimized layouts.	It may limit human creativity and require extensive training data.	Generates new design variations based on learned architectural styles.	Applied in AI-generated urban layouts for housing developments.	It could evolve into AI-powered collaborative design tools for architects.	[27, 29, 37, 38]
Feedforward Neural Networks (FNNs)	Predicting compressive strength of eco-friendly concrete (geopolymer concrete). Enhances material sustainability and construction efficiency.	Requires high-quality experimental data for training.	Processes input data in layers to predict material properties.	Used in optimizing sustainable construction materials.	Future improvements could lead to AI-driven eco-friendly material selection.	[36]
Region-Based CNN (R-CNN)	Structural defect classification and automation of conservation activities in historic buildings. Improves accuracy in anomaly detection and documentation.	High computational requirements and a need for extensive labeled image datasets.	Detects and classifies architectural defects using DL.	Employed in AI-powered heritage site restoration.	Expected advancements may allow real-time AI-driven conservation efforts.	[35]
Optimization Algorithms (Grey Wolf Optimizer - GWO)	Enhancing predictive accuracy in sustainable material strength forecasting. Optimizes material performance and reduces experimental costs.	Requires careful parameter tuning and validation.	Uses nature-inspired metaheuristic techniques to optimize predictions.	Applied in material property forecasting for eco-friendly buildings.	Future iterations may enhance predictive modeling in innovative materials.	[36]

techniques better suited for design optimization [40]. Design constraints establish boundaries by defining parameters such as dimensions, materials, construction methods, cost, and spatial considerations. While some automation

tools are being developed to translate textual constraints into digital formats, most research still relies on manually inputting these constraints for computational processing. Further studies are needed to explore GD development and

its components to improve understanding of their relationships. GD is a step-by-step process in which designers input their goals, algorithms, and constraints into the system, generating continuously adjusted models until an optimal design is achieved [27, 40].

Intelligent planning and construction have become key driving forces in the building sector. Numerous design enhancement investigations have been carried out, utilizing computational intelligence, data-driven methodologies, and optimization principles. This growing research has fueled interest in generative modeling, which autonomously produces algorithm-driven design variations, significantly minimizing human involvement and time consumption [41].

Comparable to other branches of engineering, auto-operation is progressively taking a crucial part in architecture as developments in AI optimize multiple procedures and improve design potential. AI-driven auto-operation is explored in various facets of architectural planning, encompassing digital building representation, virtual design and construction (VDC), intelligent 3D modeling, algorithmic design, computational design, efficiency assessment and simulation, workflow coordination, report creation and assessment, and digital manufacturing [42]. Regarding the latter, auto-operation revolutionizes how different architectural structures and components are conceptualized and produced. Sophisticated additive fabrication (AF) techniques, such as three-dimensional printing (3DP), enable creators to apply swift model development approaches and construct intricate and tailored elements directly, minimizing resource wastage and building duration [43]. Furthermore, auto-operation leveraging AI and 3DP is extensively studied across various domains utilizing diverse substances and fabrication scales, including metal-based 3DP methods for aviation and mechanical sectors, components produced from composite materials through photopolymer-based 3DP, along with advancements in biomedical substances via AI-driven three-dimensional and four-dimensional printing technology research [42].

AI-driven approaches, including generative adversarial networks (GANs), deep reinforcement learning (DRL), and ML, are transforming architecture by advancing adaptive facades, architectural design, structural analysis, and intelligent materials and robotics that enhance durability and performance [44]. Collectively, these studies illustrate how generative design and computational creativity are reshaping architectural workflows, opening new avenues for innovation and sustainability (as detailed in Table 2, which summarizes AI-based methods, their applications, benefits, and associated challenges).

3.3 AI in sustainable architecture and energy optimization

AI has emerged as a critical enabler in advancing sustainable architectural design, particularly through its capacity to optimize energy consumption, material selection, and environmental performance. A growing body of literature highlights how AI tools, when integrated into design workflows, support architects in simulating ecological conditions, analyzing historical energy data, and generating low-carbon material solutions [45].

ML models have been employed to analyze consumption patterns in high-rise buildings, offering predictive insights for HVAC optimization and operational efficiency. For instance, recent studies demonstrate the utility of AI in calibrating building systems based on historical occupancy and weather data, enabling more adaptive and energy-resilient designs. Tari et al. (2024) demonstrated that AI-based computational fluid dynamics (CFD) simulations could also be applied to optimize ventilation in green buildings, reducing cooling loads to similar levels [46]. Furthermore, the integration of AI with the Internet of Things (IoT) enables real-time energy regulation through dynamic sensor feedback and automated system adjustments [47]. In addition, various AI-powered tools are used to assess the carbon footprint and recyclability of construction materials [48].

Urban growth has led to increased energy demand in buildings, contributing significantly to CO₂ emissions. The construction sector accounts for nearly 30% of global energy consumption, highlighting substantial potential for efficiency improvements [49]. The building envelope is crucial for energy savings, and Building Energy Modeling (BEM) tools aid in optimizing designs. Despite advancements in predictive modeling, challenges such as inconsistencies between optimized models and architectural plans persist. Traditional design methods depend on iterative modifications, which limit efficiency [49]. To address these limitations, researchers propose hybrid strategies that integrate ML algorithms with evolutionary optimization techniques. This combination enhances the accuracy and adaptability of simulations across diverse building typologies by incorporating climatic, structural, and user behavior variables [47, 49–51]. These approaches show promise in generating context-specific energy solutions. However, several challenges hinder widespread implementation. High initial costs make it difficult for some organizations to adopt AI technologies, while the need for advanced computational infrastructure and concerns over data privacy, particularly in systems that collect and process large volumes of sensitive information, remain significant barriers [52].

Ultimately, while AI offers substantial potential to transform sustainable architectural practices, its effective

Table 2 AI-Based methods for generative design in architecture: applications, benefits, and challenges

AI-based method	Use case	Description	Key contributions	Benefits	Challenges	Ref.
Generative Adversarial Networks (GANs)	Adaptive Facades	Uses DL models to generate architectural designs that adapt to environmental conditions.	Automates facade designs for energy-efficient structures.	Energy-efficient structures, reduction in energy costs, dynamic environmental response	High computation cost, complex training process	[44]
Deep Reinforcement Learning (DRL)	Urban Planning Automation	AI-driven simulations for optimizing land use, zoning, and infrastructure.	Reduces manual effort in zoning and enhances sustainability.	Faster zoning approvals, optimized land use, enhanced urban sustainability	Complexity in execution, need for extensive training data	[45, 46]
Parametric AI Tools	Automated Blueprint Generation	Uses AI-driven rules to generate architectural blueprints and floor plans.	Improves efficiency in large-scale planning and customization.	Customizable layouts, reduction in material waste, improved workflow efficiency	Limited aesthetic intuition, high dependency on input parameters	[27, 29]
AI-Driven Structural Simulations	Earthquake-Resistant Designs	Simulates structural integrity under different conditions for risk assessment.	Enhances predictive analysis for disaster-resistant structures.	Enhanced safety standards, predictive maintenance, data-driven risk assessment	Dependence on extensive structural data, computational intensity	[30, 31]
Smart Material AI	Self-Adapting Structures	AI optimizes material properties for responsive and sustainable designs.	It helps create buildings that adapt to environmental changes.	Sustainable construction, real-time material adaptation, longevity improvement	Costly implementation, limited material database	[45, 49]
Data-Driven Optimization	AI-Enhanced Building Performance	Uses ML algorithms to optimize energy consumption, material efficiency, and maintenance schedules.	Reduces operational costs and improves efficiency.	Reduced material waste, lower lifecycle costs, enhanced operational efficiency	Requires extensive training data, interoperability challenges	[23, 24, 33]
Machine Learning (ML) Algorithms	Automated Construction Planning	AI-based models predict and optimize construction schedules and budgets.	Minimizes project delays and cost overruns.	Improved accuracy in project execution, reduction in planning errors, cost savings	Understanding AI's role in complex decision-making requires robust training datasets.	[31, 32]
Artificial Neural Networks (ANNs)	Structural Load Predictions	AI predicts load distribution for optimized architectural designs.	Improves safety and material efficiency in construction.	Improved safety, optimized materials, enhanced performance assessment	Requires extensive labeled datasets, computational power constraints, model interpretability	[24, 34]
Genetic Algorithms	Automated Space Planning	It uses evolutionary algorithms to generate optimal spatial layouts.	Enhances space efficiency in urban and interior designs.	Space-efficient layouts, adaptive designs, reduced planning effort	High computational cost, long optimization time, and the requirement of expert tuning	[27, 40]
Ant Colony Optimization (ACO)	Structural Adaptation	AI mimics nature to create self-adaptive architectural designs.	Optimizes resource utilization and improves resilience.	Nature-inspired optimization, improved resilience, efficient material usage	Complexity in implementation and lack of standardization require interdisciplinary expertise	[41, 42]
Convolutional Neural Networks (CNNs)	Architectural Style Transfer	AI applies artistic or historical styles to architectural renderings.	Automates design aesthetics, enabling quick concept visualization.	Automated design aesthetics, improved visual realism in architectural renderings	Requires large labeled datasets, computationally expensive	[35, 36]
Evolutionary Algorithms	Design Optimization	AI iterates thousands of design variations to find the best-performing solution.	Reduces manual effort and improves sustainability.	Automated iteration of design variations, energy-efficient solutions	Long processing time, computational complexity	[29, 37]
Reinforcement Learning (RL)	Construction Robotics	AI trains robots to automate construction tasks and improve efficiency.	Enhances automation in complex construction tasks.	Improved automation in on-site construction, real-time adaptability	Data-heavy training process, risk of unpredictable outcomes	[55, 56]
AI-Driven Auto-Operation	Digital Building Representation, VDC, 3D Modeling	AI optimizes multiple architectural planning processes, including workflow coordination, computational design, and efficiency assessment.	Enhances automation in architectural workflows and structural optimization.	Improved design efficiency, reduced waste, faster modeling, and production	High computational cost, need for domain expertise, and challenges in interoperability	[57, 58]

application depends on addressing infrastructural, regulatory, and methodological constraints. The reviewed literature underscores the necessity of continued research to refine these technologies and support their integration into mainstream architectural workflows [52].

Table 3 below illustrates various AI-driven applications in sustainable architecture and energy optimization, outlining the technologies employed, associated benefits, and implementation challenges.

3.4 AI in urban design and smart city planning

Urban AI refers to integrating AI into city environments, building on past technological advancements, and aligning with the smart city agenda. AI-driven systems are transforming urban life through applications such as intelligent assistants, autonomous vehicles, AI-powered travel, and surveillance. These technologies redefine how people interact with spaces and raise ethical concerns about privacy, democracy, and governance [53].

With its capacity for data-driven decision-making, AI is revolutionizing urban planning by enabling smart city infrastructure development. Advanced tools such as big data analytics, Geographic Information System (GIS) mapping, and AI-based predictive models help city planners optimize land use, transportation networks, and environmental sustainability [54] (see Table 4). A study by del Campo & Manning (2024) on using AI in real-time traffic monitoring

showed DL models that can accurately predict congestion points and suggest the best road network designs. Similarly, the authors proposed the application of reinforcement learning to dynamic traffic light control that can potentially result in a 23% reduction of average travel time in a simulated urban environment [4]. Quan (2022) proposed the application of reinforcement learning and an autoencoder for the traffic light control task in an urban environment, reducing travel time by 12.72% for the optimal control strategy [55].

Over time, city development has integrated various sophisticated analytical methods, with GIS being the most commonly utilized. Other innovations, such as database management frameworks (DBMS), decision-making support tools (DSS), planning assistance platforms (PSS), and knowledge-based systems (ES), have experienced different degrees of recognition [56]. While AI and intelligent urban technologies have reshaped sectors like commerce and supply chain management, city planning has been slower in adopting these advancements.

As technology has advanced, flexible building skins have become an interdisciplinary field of study, integrating computational and AI methods to enhance adaptability. Automated systems and advanced sensors optimize ventilation and shading, reducing energy consumption. AI-driven process integration further streamlines the design workflow, promoting environmentally responsible and sustainable architectural solutions [57].

Table 3 AI-Driven sustainable architecture and energy optimization

AI application	Use case	Technology used	Benefits	Challenges	Ref.
AI-Powered Energy Simulations	Climate-responsive building designs, including optimizing orientation, glazing, and shading strategies for energy efficiency	ML, predictive modeling	Reduction in energy costs	High computational cost	[23, 24]
Innovative Heating, Ventilation, and Air Conditioning (HVAC) Control Systems	Adaptive temperature regulation based on real-time data, seasonal adjustments, and occupancy patterns to optimize comfort and energy use	IoT sensors, reinforcement learning	Enhanced occupant comfort	Requires extensive training data	[47, 49]
AI-Based Daylighting Optimization	Smart glass and shading systems that adjust based on natural light availability to reduce reliance on artificial lighting	Neural networks, sensor fusion	Reduced lighting energy demand	Cost of sensor integration	[48]
Neural Network-Based Insulation Models	Dynamic insulation systems that adapt based on climate conditions and building usage, optimizing thermal performance	DL, thermal modeling	Lower heating/cooling expenses	Requires large-scale adoption	[46, 49]
AI-Driven Carbon Footprint Analysis	Green material selection, optimization of supply chain to minimize emissions, and lifecycle analysis for long-term sustainability	AI-driven lifecycle analysis, big data	Reduced environmental impact	Limited availability of databases	[12, 14]
Automated Waste Reduction Algorithms	AI-based recycling of materials, waste sorting, and optimized waste management strategies for sustainable construction	Computer vision, robotics automation	Sustainable construction	Complex logistics and waste segregation	[50, 52]

Table 4 AI applications in urban planning and smart cities: methods, benefits, and challenges

AI method	Application	Advantages	Challenges	Ref.
DL for Traffic Prediction	Smart mobility systems	Reduced congestion, improved traffic flow, better route planning through predictive analytics	High computational requirements, real-time data processing constraints, and data privacy concerns	[54, 55]
Reinforcement Learning (RL) for Traffic Signals	Adaptive traffic light control	Optimized signal timing, reduced vehicle idle time, lower emissions, dynamic learning adaptation	Data collection complexity, potential overfitting in different traffic conditions, high computational cost	[55, 56]
AI-Based Land Use Planning	Zoning and infrastructure optimization	Efficient space utilization, improved urban design, better infrastructure planning, reduced urban sprawl.	Algorithmic bias concerns, lack of transparency in decision-making, difficulty in integrating diverse data sources	[31, 57]
Neural Networks for Pollution Monitoring	Air quality prediction and tracking	Real-time pollution tracking, better environmental policies, early warning systems for public health protection	Sensor network dependency, calibration, and maintenance costs, potential data inaccuracies due to environmental changes	[47, 49]
AI-Driven Disaster Risk Analysis	Flood, earthquake, and climate disaster prediction	Increased disaster preparedness, faster emergency response, reduced casualties, improved risk mitigation	Data reliability issues, integration with early warning systems, computational cost of large-scale simulations	[58, 59]
Smart Grid Optimization	AI-assisted energy distribution	Sustainable power management, reduced energy waste, enhanced grid resilience, predictive maintenance	Integration challenges with existing grid infrastructure, cybersecurity risks in AI-powered grids, regulatory concerns	[47, 48]
Computer Vision for Smart Surveillance	AI-driven security and monitoring	Automated anomaly detection, enhanced public safety, reduced emergency response time, real-time crime detection	Privacy concerns, ethical issues related to surveillance, risk of bias in AI detection models	[60, 61]
AI-Powered Waste Management	Intelligent waste collection and recycling optimization	Optimized waste disposal routes, reduced landfill overflow, improved recycling rates through AI sorting	Sensor accuracy issues, high implementation costs, adoption barriers in urban infrastructure	[50, 52]
Big Data Analytics	Urban infrastructure planning	Enhances decision-making with real-time insights, improves urban services efficiency, facilitates predictive analytics for city development	Data security concerns, integration with legacy systems, challenges in processing and storing massive datasets	[57, 58]
Geographic Information Systems (GIS) AI	Land use and transportation network optimization	Spatial analysis for better urban planning, environmental sustainability, and improved disaster preparedness	Requires constant data updates, high cost of implementation, challenges in integrating with real-time AI models	[12, 59]
Database Management Systems (DBMS) & AI	Urban data storage and retrieval	Efficiently manages and processes large datasets, supports real-time decision-making, enables AI-driven automation in city management	Security vulnerabilities, potential data silos, need for robust data governance frameworks.	[59, 60]
Decision Support Systems (DSS) & AI	Wise city policy and planning	It aids policymakers with data-driven insights, supports urban sustainability goals, and improves long-term urban management strategies.	It requires high-quality data, has potential bias in AI models, and has difficulty integrating qualitative and quantitative data.	[57, 58]
Planning Assistance Platforms (PSS) AI	Urban development projects	Provides strategic insights for city growth, enhances stakeholder collaboration, and supports scenario-based urban modeling	Needs integration with other AI tools, challenges in stakeholder adoption, and complexity in urban governance frameworks	[57, 58]
Knowledge-Based Systems (ES) & AI	Smart city governance	Supports automated decision-making, improves regulatory compliance, and enhances city administration efficiency	Knowledge acquisition is complex; there is a risk of outdated decision rules and challenges in AI explainability.	[56–58]

According to various research papers, AI can also be incorporated into disaster-resilient architecture to predict earthquake resistance, flood vulnerability, and wind resilience. Architects can use ML algorithms to proactively design structures that mitigate risks, thereby supporting more effective post-disaster recovery through analysis of historical data [58, 59].

Another emerging field is AI-driven material science, which develops bio-based, low-carbon, and recyclable building materials using AI recommendation systems. These innovations help reduce the environmental impact of

construction while preserving structural integrity and aesthetic quality [60].

Table 4 below illustrates the diverse applications of AI in urban planning and smart cities. It highlights the specific AI methods, their practical use cases, advantages, and the associated implementation challenges.

4 Integration of real-world case studies in AI-driven architectural practice

Although AI technologies have been comprehensively documented in terms of technical capabilities, their practical utility is most convincingly demonstrated through real-world case studies. These projects serve not only as proof-of-concept applications but also highlight the potential and limitations of AI integration within architectural workflows and urban contexts. Cases such as **The Edge** in Amsterdam illustrate the application of AI-based environmental control systems that respond in real time to user behavior and indoor conditions [61]. Similarly, **Google's Bay View Campus** in California incorporates predictive simulation to enhance solar optimization and energy efficiency [62]. **Hyundai Motorstudio Goyang** in South Korea has adopted generative AI algorithms to refine spatial articulation and structural logic [63]. In heritage conservation, the **Basilica of Santa Maria di Collemaggio** demonstrates how DL methods can classify and interpret structural damage in post-seismic contexts [64]. Notably, **Al Bahr Towers** in Abu Dhabi integrates AI into a responsive façade system that combines traditional aesthetics with environmental regulation. Taken together, these cases reflect an emerging shift toward computational methods that not only enhance performance outcomes but also challenge conventional boundaries between form, function, and sustainability [65].

5 Discussion

The integration of AI architectural design signifies a fundamental shift in design methodology [1]. As evidenced in the literature, AI is increasingly regarded not as a supplementary tool but as a central mechanism for enabling generative creativity, performance-driven optimization, and sustainable innovation. Techniques such as ML, DL, and GANs support predictive modeling and algorithmic workflows, allowing for more adaptive and data-informed design processes [1, 2].

The following discussion critically engages with the principal challenges and ethical considerations surrounding AI integration in architectural design, including concerns related to data privacy, algorithmic bias, and the evolving role of human creativity [59]. It further distinguishes between AI methodologies and conventional computational approaches, underscoring their conceptual and functional divergences. Moreover, the discussion outlines future research trajectories and strategic recommendations, emphasizing the importance of climate-responsive design, immersive technologies such as VR/AR, heritage preservation, and interdisciplinary collaboration to ensure ethically

grounded and contextually adaptive AI applications in the built environment [59].

5.1 Challenges and ethical considerations

In the rapidly evolving landscape of architectural design, the integration of AI presents both transformative opportunities and multifaceted ethical challenges. As design workflows become increasingly data-driven and algorithmically enhanced, critical questions emerge regarding data sovereignty, decision-making authority, and the future role of human creativity. To ensure responsible implementation, it is essential to address concerns, including data privacy issues, security risks, algorithmic bias, and the evolving role of human creativity in the design process [22, 66].

5.1.1 Data privacy, security, and interdisciplinary gaps

AI-driven architecture, predictive analytics, and intelligent surveillance enhance urban planning but raise privacy and security concerns. Biometric identification and facial recognition improve traffic control and security, enabling widespread surveillance. Ethical dilemmas arise as personal data often lacks clear standards for collection, storage, and use [66].

AI-assisted building automation enhances smart homes and offices but poses security risks by collecting private user data. Continuous data collection for lighting, climate, and security systems increases vulnerability to cyberattacks, potentially allowing unauthorized access to homes and businesses [56].

Consequently, AI-driven architecture should incorporate the principles of privacy by design to mitigate the risks, which involves transparent, secure, and regulated data collection. To balance innovation with ethical responsibility, robust cybersecurity protocols, encryption methods, user consent mechanisms, and other suitable measures must be prioritized [56].

The principle of “privacy by design” necessitates that data protection considerations be embedded into the architectural design process from inception rather than as retroactive safeguards. For instance, smart buildings equipped with AI-driven occupancy sensors and behavioral monitoring systems must incorporate anonymization protocols, data minimization strategies, and user consent mechanisms at the software and hardware levels [67]. A practical application can be seen in intelligent lighting systems that adjust according to movement but store only anonymized metadata, thereby eliminating the risk of personal identification. Additionally, facial recognition technologies, when deployed in urban security or access control systems, must operate under transparent governance models that clearly

define data retention periods, access rights, and regulatory oversight [67].

The effective integration of AI in architectural practice is often impeded by enduring interdisciplinary gaps. While architects are primarily educated in spatial reasoning, aesthetic judgment, and user-centered design, they frequently lack proficiency in algorithmic logic, data structures, and computational modeling, which are core competencies essential to AI development. This asymmetry in domain expertise presents a significant barrier to collaborative innovation and underscores the need for cross-disciplinary literacy and integrated design-computation pedagogies [68].

5.1.2 Algorithmic bias and decision-making in AI architecture

A significant concern in architectural design is the potential bias in AI models. Historical datasets are the training datasets AI uses to learn, and if the datasets are skewed or not representative, they will reinforce preexisting inequalities in design and accessibility [3, 4]. A critical inquiry emerging in the context of computational intelligence is whether it possesses the capacity to replicate or potentially surpass human creativity within the architectural discipline. AI demonstrates powerful capabilities in optimization, generative modeling, and structural simulation. However, despite these advancements, it remains inherently limited in its capacity to interpret cultural contexts, ethical dilemmas, and the emotional and symbolic nuances that are integral to architectural authorship [3].

5.1.3 Human creativity vs. AI-generated design

Research indicates that architectural decision-making is not purely rational, as architects often settle for satisfactory solutions rather than seeking the optimal design. Various human and social factors influence this tendency. Cognitive biases frequently shape architectural choices, prompting the introduction of structured methods to evaluate their impact. Studies have examined numerous cognitive biases concerning the contextual factors that influence architectural decisions [69].

AI-driven architecture is one of the most debated topics, particularly regarding the role of human creativity versus AI-generated design. AI shines in data-driven and optimization-driven design, generative design, and structural analysis, but falls short in the ways that the most innately human, cultural, intuitive, and emotional architects push a building to exist beyond the results of technical problem-solving [70].

Visualization of complex, highly optimized structures can be created by AI-powered tools such as parametric design

software, which can generate elaborate and optimized structures that would otherwise take human architects months to develop. While AI does the heavy lifting of these designs, these designs often lack any sense of aesthetic subjectivity or contextual understanding, and it is uncertain if AI can replace human creativity or if it only ups the quality of it [71]. Moreover, over-reliance on AI in architecture risks reducing architects to mere supervisors of algorithmic outputs rather than active creators. The human layer, though, may not be involved in the ideation of these very plausible yet untested solutions, which could have disastrous outcomes—they did not account for the historical, cultural, and environmental layers that should exist in place of the AI [71].

The fear that AI may diminish the role of architects to the point of rendering the profession obsolete and folding it into computer science is unfounded and reductive. Architecture is not solely a problem-solving enterprise driven by data or algorithms; it is a profoundly humanistic and contextual practice. Architects operate at the intersection of design, society, culture, and the environment. While AI can assist in the production and evaluation of designs, it cannot assume authorship in the way human designers can, as it lacks the consciousness, intent, and experiential knowledge necessary for creating spaces with identity, empathy, and meaning [69].

Rather than dissolving the profession, AI challenges architects to redefine their roles as creative leaders, ethical stewards, and interdisciplinary collaborators. Far from displacing architects, AI requires them to engage more deeply with technology, with human needs, and with the broader systems in which architecture operates. This evolution will not relegate architecture to a branch of computer science but will instead enrich it through synergy [69].

AI should be embraced not as a replacement for the human mind but as a generative partner that extends the boundaries of design thinking. Through thoughtful integration, architects can harness the computational power of AI while maintaining authority over conceptual vision, cultural expression, and societal relevance. This future affirms the enduring significance of architectural education and professional practice, reshaped, not erased, by the intelligent tools we develop [69].

5.1.4 Differences between AI and traditional computational design

AI and other traditional computational methods are often confused as the same, but many applications and qualities differ. Conventional computational design has supported architects by generating and modifying designs using predefined rules, parametric modeling, and deterministic algorithms. The methods listed here are strongly programmed

logic, in which the outcome is defined within the human definition and not with an independent learning or decision-making process [72].

On the other hand, AI-powered design systems utilize ML, neural networks, and generating algorithms to explore voracious datasets, recognize patterns, and develop the best feasible solutions that go contrary to the fixed rules. In contrast to conventional computational tools in which the user explicitly specifies the qualification of every design modification, AI models are trained on data, design according to contextual constraints, and autonomously create novel architectural solutions [1, 2]. The most superficial way of expressing this is in the use of generic and reinforcement learning algorithms to optimize architectural models based on AI, whose formulations define the knowledge encoded in these models and allow them to refine themselves, providing solutions that even the most creative human designer may not use due to complexity or computational handicap [1, 2]. Parametric design and energy efficiency optimization are sufficient evidence of AI's superiority over conventional computational design. AI-driven software simulates models, predicts environmental impact, analyzes user behavior, and dynamically adapts designs to improve sustainability and functionality [1]. AI-based systems can also be adaptable in real time to change structures based on live sensor data, a feature that static computational models cannot do [1, 2]. To illustrate these differences more clearly, Fig. 2 presents a

comparative overview of AI-driven and traditional architectural design methods, highlighting key distinctions in data integration, design strategies, and sustainability. AI-Driven Design scores very high (5) in areas like Data Analysis, Automation & Speed, Integration with Tech, and Emerging-Tech, showing strong performance in modern, tech-focused domains. It scores moderately (3–4) in Creativity, Ethics, and Heritage. In contrast, Traditional Methods score higher in Heritage & Restoration (4) and Ethical Considerations (4), but remain moderate to low (2–3) in most other categories, especially in tech-related aspects. This highlights AI's strength in innovation and efficiency, while traditional methods retain value in human and cultural elements.

5.2 Future research directions

As AI advances, its role in architectural design will expand far beyond its current applications, leading to innovative ways to conserve energy, reduce emissions, and enhance the creative process. Future research should focus on the following key areas to maximize AI's impact while addressing ethical and practical challenges.

5.2.1 Digital transformation and industry innovation

The Smart Industry Revolution is reshaping architecture and design in the construction industry through rapid

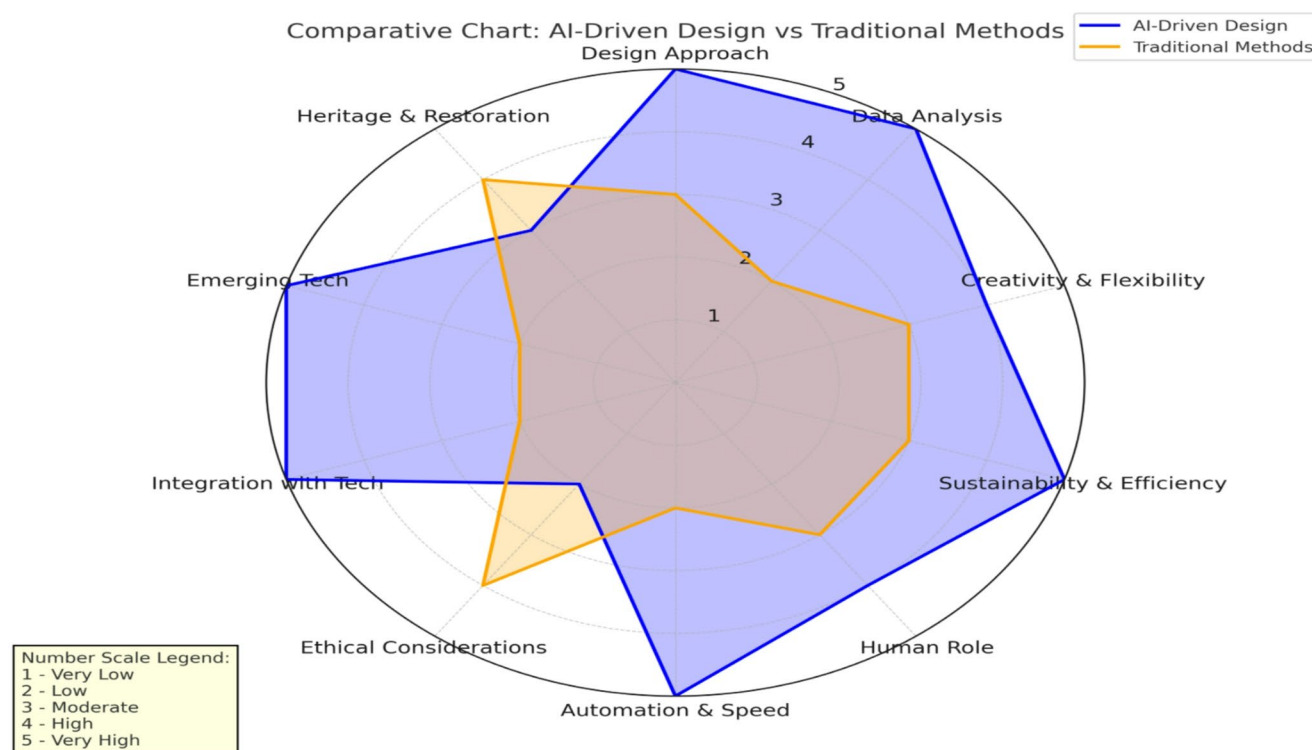


Fig. 2 A radar chart comparing AI-Driven Design and Traditional Methods across ten design criteria, using a 1–5 scale

technological advancements and the increasing availability of electronic data. The adoption of digital tools has surged, mainly due to the COVID-19 pandemic, which accelerated mobile and cloud computing use. While some companies have struggled, those embracing digital technologies, such as BIM and Digital Twin (DT), are poised to emerge stronger. These technologies help improve efficiency, sustainability, and collaboration across the design and construction phases. As the industry shifts towards these innovations, smaller firms face challenges in adopting such technologies due to cost barriers. Nonetheless, digital solutions are becoming vital for ensuring safety, monitoring progress, and facilitating construction work [15, 35, 73]. The future of architecture and construction design is centered on these transformative technologies, which are expected to drive growth, enhance project management, and improve overall outcomes. Future research will focus on identifying disruptive and emerging technologies, analyzing lifecycle management, and exploring sustainability in architectural design and construction [73].

5.2.2 Urban planning and digital twins

Urban planning, as another frontier of AI application, is transitioning toward real-time responsiveness. While smart cities have incorporated elements of AI, such as traffic prediction and energy load balancing, more comprehensive models, particularly those using digital twins, are still emerging. Future studies must investigate the role of AI in predictive zoning, multi-objective land-use planning, and infrastructure forecasting [15, 35]. The urban twin concept should evolve from static replication to dynamic prediction models that simulate behavioral, environmental, and infrastructural changes. This can support policymakers in evaluating urban interventions before implementation. However, success will depend on the availability of interoperable urban datasets and cross-sector collaboration between data scientists, municipal authorities, and architectural designers [73].

5.2.3 Climate-responsive and disaster-resilient architecture

All AI research should emphasize climate-responsive and disaster-resilient architecture to support environmentally responsible urban development [59].

Architecture integrates innovative solutions like adaptive building façades to address climate change and sustainability. These intelligent systems regulate temperature, ventilation, and solar radiation, enhancing energy efficiency and comfort through innovative materials and automation. Just as AI plays a central role in various climate-responsive design practices, such as assessing weather patterns,

optimizing energy efficiency, and promoting sustainable materials, it can also contribute to advancing climate-responsive design (Table 5). Future research should focus on AI-driven parametric design tools that generate structures based on real-time climate data. These tools would allow architects to develop self-adjusting facades, passive cooling systems, and solar-optimized layouts with little or no energy consumption [58].

With the intensifying effects of climate change and the global call for decarbonized construction, climate-responsive and disaster-resilient architecture is no longer a design ambition but a necessary standard. AI systems capable of simulating environmental variables in real time offer promising avenues for optimizing building performance in response to dynamic climatic conditions [58]. Research is required to explore how reinforcement learning, when applied to structural adaptation models, can predict and respond to extreme weather events, energy fluctuations, or material stress. These technologies must be integrated with parametric design tools that allow architectural elements to adapt autonomously. However, such tools will require an ecosystem of high-fidelity environmental data, responsive feedback loops, and domain-specific calibration strategies, areas that remain underdeveloped [58, 59].

5.2.4 AI, VR, and AR in immersive design

With AI, VR, and Augmented Reality (AR) integration, architectural visualization and design processes can potentially have a new lease of life. VR and AR technologies enable architects, clients, and stakeholders to experience and interact with existing and novel designs in immersive environments augmented with real-time data and video feeds. The following research step will determine how AI can improve real-time simulating models within VR and AR platforms so architects can experiment with design variations in real time [74]. These could be generative AI-powered models that change designs per user preferences, environment considerations, and structural constraints. In addition, the existing progress in haptic feedback and foresighted AI exploration could somehow boost the practicality of architectural walkthroughs through VR. There is another opportunity to explore using AI-enabled AR for on-site construction assistance. AR interfaces, augmented by AI, could add digital models over construction, helping architects and engineers see through the physical design, catch inconsistencies, ensure spatial accuracy, and cut material waste [74].

In parallel, the accelerated development of Extended Reality (XR), encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), has introduced new modalities of architectural visualization. Yet, the research community has not fully addressed the potential

Table 5 Future directions of AI in architecture and construction

Research focus	Key applications	Potential benefits	Challenges & Considerations	Future research directions	Ref.
AI, VR, and AR in Architecture	- AI-powered generative design in VR/AR - Real-time design simulations - AI-assisted spatial planning - AR-guided construction workflows	- Improved design visualization and iteration - Reduced construction errors - Enhanced collaboration and client interaction	- High computational costs - Integration with existing workflows - Ethical concerns in AI-generated aesthetics	- Advancing AI-driven generative models for adaptive design - Enhancing real-time AR overlays for construction validation - AI-driven automated code compliance verification for VR-based designs	[62, 74]
AI-Assisted Renovation & Heritage Conservation	- AI-driven material analysis for aging structures - Automated damage assessment via AI-powered drones - AI-assisted restoration planning	- Preserving cultural heritage - Reducing restoration costs - Predicting long-term structural integrity	- Ensuring AI-generated reconstructions remain historically accurate - Ethical concerns in altering heritage sites	- Developing AI models trained on diverse historical architecture datasets - Integrating AI with 3D printing for precise restoration - AI-driven predictive algorithms for climate impact on historical sites	[59, 64]
AI's Role in Climate-Responsive & Resilient Architecture	- AI-driven parametric climate modeling - Self-regulating building facades - AI-powered disaster resilience simulations	- Enhanced energy efficiency - Reduced carbon footprint - Better disaster preparedness and mitigation	- Limited availability of high-quality climate data - Balancing sustainability with cost-effectiveness	- Advancing AI tools for climate-adaptive architecture - AI-assisted selection of bio-based, recyclable materials - AI-integrated urban heat island effect mitigation strategies	[45, 49]
AI for Smart Cities and Urban Planning	- AI-optimized zoning and land use analysis - AI-powered traffic flow modeling - Predictive maintenance for urban infrastructure	- Reduced urban congestion - More efficient use of resources - Improved public safety and sustainability	- Managing large-scale urban data sets - Ethical concerns in AI-driven decision-making	- Integrating AI with IoT for real-time urban monitoring - AI-assisted green city planning with renewable energy - AI-driven simulation models for optimizing urban resilience against extreme weather	[54, 55]
AI in Sustainable Construction Materials	- AI-optimized selection of low-carbon materials - AI-driven material lifecycle analysis - AI-assisted 3D printing of eco-friendly structures	- Reduced construction waste - Increased durability of materials - Cost-effective and sustainable building practices	- Limited adoption of AI in traditional construction - Need for regulatory approval of AI-designed materials	- AI-based material science for biodegradable and self-repairing materials - AI-powered predictive analysis for long-term durability - AI-driven structural optimization for reducing embodied carbon	[49, 52]

of merging XR platforms with generative AI systems [74]. Future studies must examine the extent to which AI can support real-time spatial manipulation in immersive environments. In this context, generative algorithms could act as co-design agents, interpreting verbal, visual, or sensory inputs from users to iteratively refine design proposals. When further complemented by haptic feedback and AI-informed material simulations, the result could be a fully

interactive design experience — a concept that remains largely theoretical but is urgently worth investigating [74].

5.2.5 Heritage conservation and AI-assisted renovation

In the context of heritage conservation, AI presents opportunities for automated damage assessment, predictive aging models, and data-informed restoration planning—yet further inquiry is needed to ensure historical fidelity and

ethical intervention [35]. Climate-resilient architecture also remains underexplored, particularly concerning AI-driven adaptive façades, passive system optimization, and real-time risk modeling for natural disasters. Additionally, developing open-access datasets and interoperable platforms would enhance the practical viability of AI tools in architectural practice. Addressing these gaps requires cross-disciplinary frameworks that bring together designers, data scientists, historians, and environmental engineers to collaboratively shape a technologically advanced yet culturally responsive built environment [59].

Structural deterioration, material limitations, and the necessity of maintaining authenticity in restoring these buildings and cultural heritage sites present particular preservation challenges. In such cases, AI can revolutionize heritage conservation by analyzing architectural styles, predicting material aging, and creating restoration plans based on the first architectural style with modern safety standards while retaining structural integrity [35].

Table 5 below outlines the future research directions, benefits, applications, and challenges associated with AI integration across architecture and construction subfields, including heritage conservation, climate-responsive design, smart cities, and sustainable materials.

5.2.6 Intelligent materials and predictive modeling

From a materials perspective, the emergence of bio-based and recyclable composites has broadened the possibilities for sustainable construction. However, these materials often lack robust data on long-term performance under varied stress conditions. AI has the capacity to bridge this gap through predictive modeling [1]. Future research should address the integration of AI in evaluating lifecycle performance, toxicity levels, and thermal efficiency of next-generation construction materials. Optimization algorithms such as genetic programming and swarm intelligence could assist in identifying the ideal composition of hybrid materials [1, 2]. This, however, demands a multidisciplinary research model where material scientists, architects, and computer engineers operate within a unified computational framework.

5.2.7 Governance, education, and ethical frameworks

Lastly, governance, education, and public trust will play a decisive role in AI's architectural future. The professional divide between architects and technologists can no longer be sustained. It must be acknowledged that the introduction of AI into design practice necessitates a recalibration of pedagogical content across architectural institutions [75]. Future curricula should incorporate algorithmic thinking,

ethical reasoning, and systems literacy to prepare students not just to use AI, but to critique and shape it. Likewise, the research community must focus on the development of regulatory protocols that ensure transparency, inclusiveness, and accountability in AI-generated design decisions. Without such frameworks, the integration of AI into architecture risks being driven by market dynamics rather than public benefit [75–78].

5.3 Limitations

Despite adhering to a structured and rigorous review methodology, this study is subject to certain limitations that warrant consideration. The exclusion of non-English literature may have inadvertently limited the scope of the review by omitting regionally significant developments and culturally specific implementations of AI in architecture. Furthermore, the reliance on established academic databases such as Web of Science, Scopus, and Google Scholar, while ensuring the inclusion of high-impact peer-reviewed sources, may have resulted in the omission of valuable insights published in niche journals, conference proceedings, or industry-specific repositories not indexed in these platforms. As such, the synthesized findings, although reflective of prevailing academic discourse, may not fully encapsulate the diversity of global practices or emerging perspectives situated outside mainstream publication channels.

6 Conclusion and recommendations

This research highlights a promising avenue for future work: developing AI algorithms trained on historical architectural data. These models can help reconstruct missing design elements, recommend sustainable restoration materials, and simulate environmental impacts on heritage buildings. AI-powered drones and robotic scanning systems can produce high-resolution 3D maps of endangered sites, supporting conservationists in damage assessment and preservation planning.

However, such research must address the ethical considerations of using AI to recreate heritage. While AI can improve restoration accuracy, interventions must be guided by cultural integrity and historical authenticity rather than modern biases or stylistic assumptions.

AI in architecture enhances efficiency, creativity, and sustainability by optimizing workflows and structures. To maximize these benefits and mitigate risks, AI models must be transparent, unbiased, and inclusive. Policymakers and industry leaders should establish ethical guidelines to prevent architectural homogeneity, exclusion, and privacy concerns. AI-driven designs must respect diverse cultural and

social needs, upholding architectural integrity and ensuring inclusivity.

Importantly, AI should not be seen as a replacement for human architects, but rather as a creative partner that augments their work. Incorporating AI training into architectural education will help future professionals understand how to use AI tools effectively while respecting artistic and cultural values.

All AI research should emphasize climate-responsive and disaster-resilient architecture to promote environmentally responsible urban development. AI-based solutions can improve energy efficiency, optimize material usage, and support the design of sustainable cities, ultimately reducing the environmental impact of architectural projects.

Future work should also focus on integrating AI with emerging technologies such as VR, AR, and robotics. These combined technologies can create more immersive, efficient, and innovative design processes, transforming architectural design into a dynamic canvas for creativity. Finally, strong security and privacy frameworks are essential for AI-driven smart buildings. The future of AI in architecture depends on its ethical integration, enhancing design, and enriching the built environment.

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Declarations

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